

# Alex An

647-327-0031 | [alexan0406@gmail.com](mailto:alexan0406@gmail.com) | [linkedin.com/alex-an](https://linkedin.com/alex-an) | [github.com/alex-an0406](https://github.com/alex-an0406)

## EDUCATION

### Bachelor of Applied Science in Computer Engineering

University of Toronto

Toronto, ON

Minor in Artificial Intelligence

Expected Graduation: April 2028 + PEY

**Relevant Coursework:** Software Engineering, Data Structures and Algorithms, Software Design & Communication, Computer Networks, Computer Organization

## EXPERIENCE

### BuyCAN - [buycan.github.io](https://buycan.github.io)

Feb 2024 – Present

Software Developer

Toronto, ON

- Grew a consumer shopping application to 6,000+ active downloads and a top 100 ranking on Apple's shopping apps chart by leading backend and frontend architecture
- Cut releases from two separately maintained native codebases to one iOS & Android deploy by migrating Swift and Kotlin code into React Native and implementing complete CI/CD pipelines
- Served every client from one backend under concurrent asynchronous load by building a scalable Python API server with RESTful endpoints and secure JSON payloads
- Reduced the cost of an LLM research pipeline (OpenAI, Perplexity, Google Search) by optimizing model calls, adding a call to cost logging system, and routing tasks to cheaper models where possible

### AI Annotator

Jan 2025 – Present

Outlier AI

Remote

- Improved the accuracy and reasoning of large language models (LLMs) by annotating multi-modal text, image and code datasets against strict rubrics and debugging AI generated code during quality assurance review

## PROJECTS

### CourtFinder Full-Stack App - [courtfinder.app](https://courtfinder.app) | TypeScript, Next.js, PostGIS, Supabase

- Shipped a full-stack web application on Vercel for locating public tennis and pickleball courts and reporting live availability status/conditions
- Designed a PostGIS database on Supabase using row-level security, database rate limits and triggers to safely accept anonymous edits from the browser
- Enforced a trust model requiring 3 independent confirmations to approve user submissions through Postgres triggers and 24 row-level security policies

### Geographic Information System (GIS) & Router | C++, Data Structures, Git

- Architected a mapping and route planning software over OpenStreetMap datasets of 1,000,000+ street segments
- Optimized paths across a 100,000+ segment weighted graph while respecting one way and turn restrictions by applying A\* search and Dijkstra's algorithm with priority queues
- Solidified the system with unit tests across spatial queries, routing logic and API clients in a team of 3 using Git, feature branching, code review and CMake builds on Linux
- Solved the Traveling Salesman Problem in a hard 50 second runtime cutoff by combining distance precomputation, greedy construction and heuristic optimization
- Placed 4th overall in a cohort of 300+ engineering students measured through performance against 12 benchmark scenarios

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, Swift, Kotlin, C/C++, SQL, HTML/CSS

**Frontend & Mobile:** React, React Native, Next.js, Tailwind CSS

**Backend & Data:** REST APIs, JSON, HTTP, PostgreSQL, PostGIS, SQLite, Supabase, Node.js, Flask, Azure

**Tools & Concepts:** Git, GitHub, CI/CD, Vercel, Linux/Unix, Agile